anesthesia research

MDPI

Review

Sepsis Biomarkers: What Surgeons Need to Know

Gabriele Melegari [1,*], Federica Arturi [2], Fabio Gazzotti [1], Matteo Villani [3], Elisabetta Bertellini [4] and Alberto Barbieri [2,*]

[1] Department of Anaesthesia and Intensive Care, Azienda Ospedaliero Universitaria Policlinico di Modena, 41121 Modena, Italy; gazzottifabio@gmail.com
[2] School of Anesthesia and Intensive Care, University of Modena and Reggio Emilia, 41125 Modena, Italy; federica.arturi.94@gmail.com
[3] Department of Anaesthesia and Intensive Care, Azienda Ospedaliero Universitaria Parma, 56126 Pisa, Italy; mvillani@live.it
[4] Faculty of Medicine, University of Modena and Reggio Emilia, 41121 Modena, Italy; elisabettab53@gmail.com
* Correspondence: melegari.gabriele@aou.mo.it or melegari.gabriele@gmail.com (G.M.); alberto.barbieri@unimore.it (A.B.)

Abstract

Background: Sepsis is a life-threatening syndrome caused by a dysregulated host response to infection leading to organ dysfunction. Distinguishing sepsis from localized infection is crucial, as it guides clinical decision-making and biomarker interpretation. Biomarkers may support diagnosis, prognosis, and therapeutic choices, but their integration into practice remains debated. Methods: This narrative review was conducted in accordance with the SANRA (Scale for the Assessment of Narrative Review Articles) guidelines. A comprehensive literature search was performed in PubMed, Embase, and Cochrane CENTRAL (January 2000–September 2025). Studies evaluating sepsis-related biomarkers for diagnosis, prognostication, shock assessment, antimicrobial stewardship, and post-acute follow-up were considered. Findings: Established biomarkers such as procalcitonin (PCT), C-reactive protein (CRP), and lactate remain widely used for diagnosis, monitoring of inflammatory response, and assessment of severity. Emerging candidates include pancreatic stone protein (PSP), neutrophil gelatinase-associated lipocalin (NGAL), and monocyte HLA-DR (mHLA-DR), which may provide insights into infection dynamics, renal injury, and immune suppression, respectively. However, limitations in standardization and heterogeneous evidence hinder routine implementation. Interleukin-6 (IL-6), despite extensive study, shows limited specificity and inconsistent clinical applicability. Renin has been proposed as a marker of shock severity rather than infection. Comparative evidence highlights the need for stage-specific biomarker use across prehospital, emergency, ICU, and recovery phases. Conclusions: No single biomarker is universally applicable in sepsis. Their utility depends on timing, clinical setting, and patient phenotype. Combining classical and emerging biomarkers with point-of-care technologies and dynamic monitoring may enhance personalized management. Limitations include heterogeneity of evidence and lack of standardized thresholds. Future research should validate biomarker panels, integrate them into stewardship strategies, and explore their cost-effectiveness in clinical practice.

Keywords: sepsis; biomarkers; procalcitonin; C-reactive protein; lactate; pancreatic stone protein; mHLA-DR; NGAL; renin; antibiotic stewardship

check for updates

Academic Editor: Thierno Madjou Bah

Received: 29 July 2025
Revised: 16 September 2025
Accepted: 2 October 2025
Published: 13 October 2025

Citation: Melegari, G.; Arturi, F.; Gazzotti, F.; Villani, M.; Bertellini, E.; Barbieri, A. Sepsis Biomarkers: What Surgeons Need to Know. *Anesth. Res.* **2025**, *2*, 23. https://doi.org/10.3390/anesthres2040023

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

Personalized medicine, also known as precision medicine, is a medical approach that customizes disease prevention, diagnosis, and treatment to individual patients based on their genetic and genomic information. This concept has particular relevance in complex conditions such as sepsis [1,2]. Sepsis is currently defined as a life-threatening condition caused by a dysregulated host response to infection leading to organ dysfunction [3]. It is important to underline that sepsis must be clearly distinguished from infection: while infection refers to the presence of a pathogen that triggers an immune response which may remain localized, sepsis represents a maladaptive systemic reaction that causes tissue damage, immune dysregulation, and ultimately organ failure [4].This distinction is crucial for interpreting biomarker profiles and for making appropriate therapeutic decisions.

Biomarkers play a central role in sepsis not only for early diagnosis but also for differentiating infectious from non-infectious inflammation, for prognosis, and for guiding therapeutic choices. Some biomarkers, such as procalcitonin (PCT), can be used to "rule out" bacterial infection and thus reduce unnecessary antibiotic exposure, as shown by Christ-Crain et al. [5,6] finding later confirmed in stewardship programs [7]. Classical biomarkers including PCT, C-reactive protein (CRP), and lactate remain widely used in clinical practice, but emerging molecules such as pancreatic stone protein (PSP), neutrophil gelatinase-associated lipocalin (NGAL), and monocyte human leukocyte antigen–DR (mHLA-DR) expression are attracting increasing interest. These markers offer potential insights into infection dynamics, risk of acute kidney injury, and immune suppression, respectively [8,9]. However, their clinical integration remains uncertain, as evidence is often heterogeneous and thresholds are not standardized. For example, interleukin-6 (IL-6) has been extensively studied but its clinical applicability is limited by low specificity and high biological variability. The purpose of this article is to provide a narrative review, conducted in accordance with the SANRA (Scale for the Assessment of Narrative Review Articles) guidelines [10]. This approach was chosen because it allows the integration of heterogeneous data from observational studies, randomized trials, and translational research, which is particularly important in sepsis, where biomarker research is fragmented and evolving. To give structure and clarity, the research question was formulated according to the PICO framework: the population of interest includes patients with suspected or confirmed sepsis or septic shock across different settings (prehospital, emergency department, hospital admission, intensive care unit, post-acute care), the intervention is the measurement of sepsis-related biomarkers, the comparator is clinical assessment or alternative markers, and the outcomes of interest are diagnostic and prognostic accuracy, prediction of complications such as shock or acute kidney injury, and the contribution to antimicrobial stewardship strategies. In order to contextualize the role of biomarkers, it is useful to recall the conceptual models described by Aronson et al. which classify biomarkers according to their relationship with disease mechanisms [11]. Table 1 summarizes these models and applies them to sepsis, showing how some biomarkers such as lactate can be considered prognostic, while others, such as PCT, are diagnostic, and still others, may serve as monitoring tools for immune function.

The development and validation of dependable sepsis biomarkers could significantly shorten diagnosis delays, reduce inappropriate treatment, and enhance patient management. This narrative review therefore provides a critical appraisal of the current landscape of sepsis biomarkers, with a focus on diagnosis, severity assessment in shock, antimicrobial stewardship, and post-acute follow-up. By comparing established and emerging markers, we aim to highlight the strongest evidence for clinical use, identify limitations, and suggest directions for future research.

Table 1. Summary of biomarker models described by Aronson and Ferner [11].

Model	Impact and Application
A biomarker directly linked to the disease mechanism, accurately predicting clinical outcomes.	This is the gold standard in biomarker research, providing a clear correlation between biomarker levels and disease progression or therapeutic response.
The biomarker is linked to an intermediate stage of the disease but does not directly predict the outcome.	This scenario is common in cardiovascular biomarkers where surrogate endpoints like blood pressure are used instead of long-term cardiovascular mortality.
The biomarker is associated with the disease but not causally linked.	While useful for risk stratification, it may not be reliable for guiding therapeutic interventions.
The biomarker correlates with the disease purely by coincidence rather a mechanistic link.	This can lead to misleading conclusions in clinical trials, necessitating rigorous validation.
A biomarker is believed to predict a clinical outcome but fails in real-world validation.	It happened frequently in observational studies. A historical example is the Cardiac Arrhythmia Suppression Trial (CAST), where antiarrhythmic drugs reduced arrhythmias (biomarker) but increased mortality (clinical endpoint), demonstrating the risk of using unreliable surrogates.

The development and validation of reliable sepsis biomarkers could significantly reduce delays in diagnosis, minimize inappropriate antibiotic exposure, and improve the overall management of patients with sepsis [4]. This perspective article provides an overview of the current landscape of biomarkers in sepsis, exploring the diagnosis, the severity assessment in case of shock, and the post-acute phase. This review also includes considerations about the settings and type of patients, including prehospital. The diagnosis of sepsis could take place in different settings such as hospital admission, intrahospital length of stay, intensive care unit (ICU), and recently also at home [12]. In each of these settings, the focus could vary from the diagnosis, following shock assessment, and sepsis recurrence. The modern era of personalized sepsis medicine should consider all these aspects, as well as the concept of the phenotypes of septic patients [13,14].

The development and validation of dependable sepsis biomarkers could significantly shorten diagnosis delays, reduce inappropriate antibiotic exposure, and enhance the overall management of sepsis patients [4]. This perspective article offers an overview of the current landscape of biomarkers in sepsis, focusing on diagnosis, severity assessment in cases of shock, and the post-acute phase. The review also considers various settings and types of patients, including those in prehospital situations. Diagnosis of sepsis can occur in several environments, such as during hospital admission, throughout the intrahospital length of stay, in the intensive care unit (ICU), and, more recently, even at home [12]. In each of these contexts, the emphasis may shift among diagnosis, shock assessment, and the recurrence of sepsis. The modern approach to personalized sepsis medicine should take all of these factors into account, including the varying phenotypes of septic patients [13,14].

2. Materials and Methods

This article is a narrative review, conducted according to the Scale for the Assessment of Narrative Review Articles (SANRA) guidelines to ensure methodological rigor.

A comprehensive literature search was carried out in PubMed, Embase, and Cochrane CENTRAL, covering the period January 2000 to September 2025. The search strategy combined the terms “sepsis biomarkers,” “septic shock,” “diagnosis,” “prognosis,” “antibiotic stewardship,” and “post-acute care”, with appropriate Boolean operators adapted for each database.

Inclusion criteria were observational or interventional clinical studies, systematic reviews, meta-analyses, or narrative reviews relevant to sepsis biomarkers; adult or pe-

diatric patients with suspected or confirmed sepsis or septic shock; outcomes addressing diagnostic, prognostic, or monitoring roles of biomarkers; and articles written in English. Exclusion criteria included case reports, conference abstracts without full data, experimental studies not involving patients, and articles not primarily focused on sepsis biomarkers. The screening process included title/abstract evaluation followed by full-text assessment to ensure relevance to the objectives of this review. A comprehensive literature search was carried out in PubMed, Embase, and Cochrane CENTRAL, covering the period January 2000 to September 2025. The search strategy combined the terms "sepsis biomarkers," "septic shock," "diagnosis," "prognosis," "antibiotic stewardship," and "post-acute care," with appropriate Boolean operators adapted for each database. The search retrieved a total of 14,251 records. After removal of 3251 duplicates, 11,000 unique articles remained. Titles and abstracts were screened, and 10,200 records were excluded following the authors' predefined criteria based on Boolean combinations, which allowed us to refine the search to studies specifically addressing sepsis-related biomarkers in clinical settings. A total of 800 full-text articles were then assessed for eligibility. Ultimately, 75 articles met the inclusion criteria and were included in the final narrative synthesis. The detailed selection process is illustrated in Figure 1.

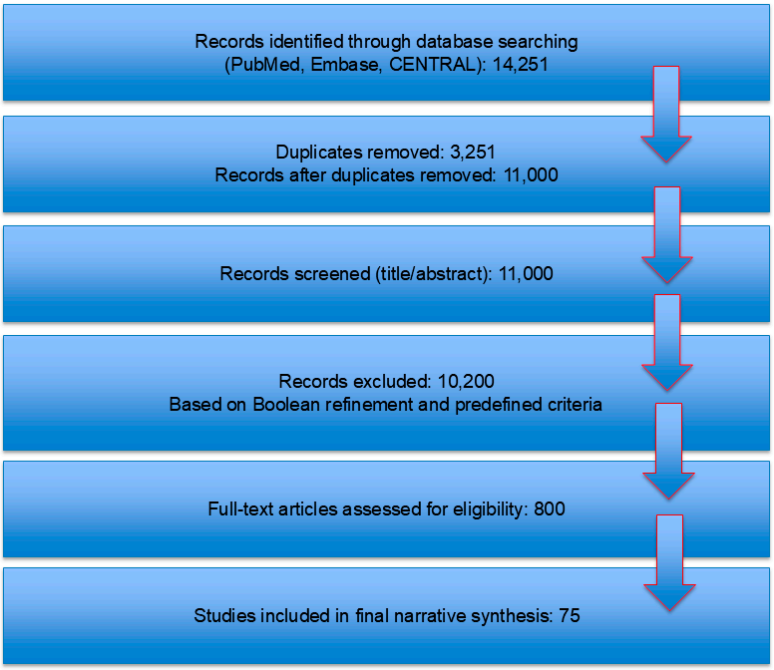

Figure 1. A descriptive flow diagram of the literature search and selection process is provided in the figure.

3. Sepsis Diagnosis Biomarkers

The initial diagnosis of sepsis typically occurs upon hospital admission, using clinical criteria, laboratory biomarkers, and established scoring systems. Sepsis guidelines recommend several biomarkers for the early detection of sepsis, including procalcitonin (PCT) and C-reactive protein (CRP). These biomarkers help assess the severity of infection and the level of systemic inflammation [15,16]. Lactate measurement is also a key component of sepsis evaluation, as elevated lactate levels (>2 mmol/L) indicate tissue hypoperfusion and are associated with worse outcomes. This guides early resuscitation strategies. Often, this biomarker is the most readily available in all settings and provides a rapid response [17]. Sepsis severity is still assessed using standardized scoring systems such as the Sequential Organ Failure Assessment (SOFA) and quick SOFA (qSOFA), which help predict mortality risk and guide clinical decision-making; however, qSOFA is not

recommended as a unique score [15]. Procalcitonin (PCT) is the most diffuse biomarker, which rises in response to bacterial infections, distinguishing sepsis from non-infectious inflammatory states [18]. Sepsis is not equivalent to infection; sepsis is a life-threatening condition, where microbiological culture is a fundamental component of the diagnosis. In particular, performing rapid antimicrobial susceptibility testing (AST) remains a clinical challenge for microbiologists and one of the most interesting research directions for the future, but at the same time the technique can only identify a limited number of pathogens present in the patient's blood [19].

Research indicates that incorporating PCT measurements can improve early sepsis prediction beyond clinical assessment alone. In the last decade, several sepsis biomarkers have been introduced, showing a high level of accuracy: presepsin, a soluble CD14 subtype, and Mid-regional pro-adrenomedullin have demonstrated potential in the early diagnosis of sepsis [20]. Studies suggest that presepsin levels correlate with sepsis severity and may serve as a reliable diagnostic marker, especially in pediatric settings [21,22]. Another vital biomarker is interleukin-6 (IL-6), a pro-inflammatory cytokine that increases rapidly in response to infection and correlates with disease severity. Anyway, it is important to underline how this biomarker signals the involvement of the vascular endothelium during cytokine storms [23]. Despite being one of the most extensively studied cytokines in relation to sepsis, interleukin-6 (IL-6) has several limitations that reduce its reliability as a standalone biomarker for infection or sepsis. IL-6 is a pleiotropic pro-inflammatory cytokine that is rapidly released in response to infection or tissue injury, causing its levels to rise early during the inflammatory response [24,25]. However, this early and non-specific reaction poses a significant drawback. IL-6 can be significantly elevated in various non-infectious conditions, such as trauma, surgery, burns, pancreatitis, and autoimmune disorders, which makes it poorly specific for diagnosing sepsis [26,27].

IL-6 levels fluctuate rapidly, have a short half-life, and exhibit significant variability between individuals, which complicates the interpretation of measurements taken at a single point in time. The absence of standardized thresholds, high biological variability, and poor correlation with clinical severity scores like SOFA or APACHE II in certain populations further diminish its usefulness in guiding clinical decisions. While some studies have linked IL-6 to outcomes such as mortality or progression to septic shock, these associations are inconsistent across different populations and study designs [28]. IL-6 lacks validated clinical algorithms for guiding the initiation or discontinuation of antibiotic therapy, unlike more established biomarkers such as procalcitonin. Moreover, assay variability and prolonged turnaround times further limit its applicability in daily practice. As a result, although IL-6 remains valuable as a research tool and for elucidating pathophysiological mechanisms, its role as a reliable and actionable biomarker in sepsis is currently marginal. At the hospital triage stage, the priority is to achieve a rapid and accessible diagnostic tool. In this context, nanotechnology-based diagnostic platforms have recently emerged as promising innovations in sepsis management. Approaches such as nanoparticle-based immunoassays, surface-enhanced Raman spectroscopy (SERS), and microfluidic nanodevices offer rapid and multiplexed biomarker detection with improved sensitivity and specificity compared with conventional assays [29]. These advancements offer significant potential for reducing time-to-diagnosis and improving early therapeutic interventions, ultimately enhancing patient outcomes in sepsis care, at the same time, this technology is still expensive and not often available. A promising solution is offered by Pancreatic Stone Protein (PSP) that has emerged as a hopeful indicator among the various biomarkers under investigation. PSP is a 14-kDa glycoprotein primarily produced by pancreatic cells, although its expression has also been observed in other tissues. Recent studies have highlighted that PSP levels significantly increase in response to systemic stress and infections, making it a potential

biomarker for sepsis. A literature review suggests that PSP exhibits superior diagnostic performance compared to traditional biomarkers such as C-reactive protein (CRP) and procalcitonin (PCT) in the early identification of sepsis [30–32]. Introducing point-of-care (POC) devices for measuring biomarkers has significantly improved their clinical potential. These devices enable rapid and accurate assessment of multiple biomarkers at the patient's bedside, allowing for timely therapeutic decisions [33]. Implementing POC devices for biomarkers such as PSP, PCT, could enable healthcare providers to identify septic patients early during home visits, allowing for the immediate initiation of appropriate therapies and consequently improving clinical outcomes [34,35]. This strategy could be particularly advantageous in home-based sepsis treatment programs, reducing the need for hospitalization and optimizing healthcare resource allocation, and it has shown interesting results in the case of pediatric septic children's patients with high accuracy in discriminating septic shock [36,37]. However, despite promising evidence, further research is needed to validate the effectiveness of multiple biomarkers as diagnostic and prognostic tools for sepsis for these last biomarkers. Future studies should focus on standardizing reference values, evaluating potential interferences, and analyzing the cost-effectiveness of implementing POC devices for these biomarkers in routine clinical practice. Table 2 provides a comparative overview of established and emerging sepsis biomarkers, summarizing their mechanisms, kinetics, clinical applications, limitations, and strength of evidence. This structured comparison aims to support clinicians in interpreting their relative utility across different stages of sepsis management. Despite their widespread use, classical biomarkers have important limitations. C-reactive protein (CRP) is inexpensive and universally available but has low specificity and delayed kinetics, which limit its role as an early marker [28]. Procalcitonin (PCT) is supported by multiple randomized trials and meta-analyses showing utility in antibiotic stewardship, yet its diagnostic accuracy remains imperfect, with false positives in trauma, surgery, and renal dysfunction. Lactate, while a strong prognostic indicator of circulatory failure, is not sepsis-specific and should be interpreted in the broader context of tissue perfusion and resuscitation [38]. Comparative studies highlight that no single biomarker can reliably distinguish sepsis from non-infectious inflammation, and area under the curve (AUC) values rarely exceed 0.80, indicating only moderate discriminative capacity. Therefore, in current practice, these markers should be considered adjunctive tools that complement, but cannot replace, clinical judgment and microbiological confirmation.

Table 2. Comparative summary of established and emerging sepsis biomarkers.

Biomarker	Mechanism/Source	Kinetics	Main Clinical Role	Limitations	Strength of Evidence
Procalcitonin (PCT)	Prohormone of calcitonin; rises with bacterial infection	Increases within 6–12 h, half-life ~24 h	Diagnosis of bacterial infection; antibiotic stewardship (initiation/discontinuation)	False positives in trauma, surgery, renal failure; limited availability in some centers	Strong: multiple RCTs, meta-analyses, stewardship guidelines
C-reactive protein (CRP)	Acute-phase protein from liver	Peaks 24–48 h, half-life 19 h	Non-specific marker of inflammation; supportive for infection/sepsis	Slow kinetics, low specificity	Strong: widely available, but low discriminative power
Lactate	Product of anaerobic metabolism; marker of tissue hypoperfusion	Rapid rise with shock; clearance within hours if resuscitation effective	Severity of circulatory failure, prognosis, resuscitation target	Elevated also in seizures, liver disease, beta-agonist use	Strong: guideline-recommended, prognostic value confirmed

Table 2. *Cont.*

Biomarker	Mechanism/Source	Kinetics	Main Clinical Role	Limitations	Strength of Evidence
Pancreatic Stone Protein (PSP)	Glycoprotein secreted mainly by pancreas; rises with systemic stress/infection	Rapid increase during infection and systemic stress	Early diagnosis of infection/sepsis; point-of-care potential	Limited validation; assays not standardized; cost	Moderate: promising evidence, some multicenter studies
Neutrophil Gelatinase-Associated Lipocalin (NGAL)	Glycoprotein from neutrophils and renal tubular cells	Early rise (hours) in kidney stress/injury	Prediction of acute kidney injury (AKI) in sepsis	Non-specific; also elevated in other renal/inflammatory conditions	Moderate: validated in AKI, limited sepsis-specific data
Monocyte HLA-DR (mHLA-DR)	Surface expression marker of monocyte antigen presentation	Decrease reflects immune suppression; recovery correlates with immune reconstitution	Identification of immune paralysis; risk of secondary infections	Requires flow cytometry; thresholds not standardized	Moderate: prognostic, promising for immune monitoring
Interleukin-6 (IL-6)	Pro-inflammatory cytokine released early in infection/tissue injury	Peaks rapidly; short half-life	Early inflammation marker; correlates with severity	Poor specificity (elevated in trauma, surgery, burns, autoimmunity); no validated thresholds	Weak: inconsistent results; limited clinical algorithms
Renin	Hormone of RAAS; compensatory rise in vasodilatory shock	Increases rapidly with hypoperfusion/shock	Marker of shock severity, prognosis; potential for vasopressor guidance	Not infection-specific; assays less available	Emerging: pilot and ICU studies, promising but limited evidence

3.1. Septic Shock Assessment and Septic Phenotype Biomarkers

Septic shock is a leading cause of illness and death in critically ill patients. It is characterized by severe dysfunction in circulation, cellular processes, and metabolism. Identifying reliable biomarkers for septic shock is essential for early diagnosis, risk assessment, and treatment monitoring. These biomarkers are connected to the pathophysiology of shock, kidney dysfunction, immunomodulation, and the various phenotypes of sepsis. In the context of septic shock, the most widely accepted and clinically utilized biomarker of shock severity is blood lactate [39,40]. Elevated lactate levels reflect impaired tissue perfusion, mitochondrial dysfunction, and a shift toward anaerobic metabolism, all of which are hallmark features of circulatory failure in sepsis [41]. Lactate has consistently been associated with poor prognosis and is included in sepsis guidelines as a key marker for early recognition and risk stratification [42]. Lactate is a non-specific indicator that can be elevated due to various conditions not related to infection, such as seizures, liver dysfunction, or the use of beta-agonists. Recently, research has focused on the renin–angiotensin–aldosterone system (RAAS) as a potential factor in the pathophysiology of shock states. In this context, plasma renin has emerged as a promising biomarker for circulatory shock, especially in vasodilatory forms like septic shock. Elevated renin concentrations are thought to reflect the physiological compensation for these conditions: hypotension and peripheral vasodilation and may correlate with persistent hemodynamic instability. Crucially, renin levels appear to rise in response to shock-related hypoperfusion rather than infection per se, thereby distinguishing renin as a marker of shock, not of sepsis [43,44]. Several studies have shown that renin levels are associated with disease severity and mortality in patients with circulatory failure, including sepsis-related shock, suggesting a potential role in both prognosis and therapeutic monitoring [44–46]. In a study by Gleeson et al. involving a diverse population

in the ICU, it was found that renin measurement was not significantly influenced by factors such as diurnal variation, continuous renal replacement therapy, or medications. Additionally, renin was identified as a marker of tissue perfusion and was shown to be a better predictor of ICU mortality than lactate [47]. Plasma renin has been proposed as a biomarker for shock and mortality, particularly in patients receiving treatment with Angiotensin II (Ang II) [48,49]. Ang II is a potent vasoconstrictor, and its deficiency during sepsis can lead to refractory hypotension [50]. Studies have indicated that low levels of Ang II in patients may contribute to this condition [51–53]. The involvement of the RAS in septic AKI has also been studied, with evidence suggesting that angiotensin II deficiency contributes to renal hypoperfusion and ischemic damage [54,55]. The overactivation of angiotensin type 1 (AT1) receptors during inflammation can worsen kidney injury by increasing oxidative stress and causing endothelial dysfunction. By understanding how the dysregulation of the RAS interacts with kidney injury, we may be able to develop targeted therapies that help preserve renal function in cases of septic shock [56]. Acute kidney injury (AKI) is a common and serious complication of septic shock. It contributes to increased mortality rates and longer stays in the ICU. Traditional biomarkers, such as serum creatinine and blood urea nitrogen (BUN), are indicators of kidney dysfunction but are often delayed in their response. In contrast, novel biomarkers—such as neutrophil gelatinase-associated lipocalin (NGAL), kidney injury molecule-1 (KIM-1), and the combination of tissue inhibitor of metalloproteinases-2 (TIMP-2) with insulin-like growth factor-binding protein 7 (IGFBP7)—have shown promise for the early prediction of AKI in patients with sepsis [57–59]. TIMP-2 and IGFBP7, reflect early tubular stress rather than overt damage, making them valuable tools for risk stratification [60]. Septic shock is characterized by a complex interplay of pro-inflammatory and anti-inflammatory responses. Dysregulated immune function can lead to persistent inflammation, immune paralysis, or both, increasing susceptibility to secondary infections and multi-organ failure. Biomarkers that reflect immune status are crucial for identifying patients who may benefit from immunomodulatory therapies [2,61]. C-reactive protein (CRP) and procalcitonin (PCT) are commonly used inflammatory biomarkers for diagnosing and predicting outcomes in sepsis. In addition to these, newer biomarkers like soluble triggering receptor expressed on myeloid cells-1 (sTREM-1), interleukin-6 (IL-6), and the expression of human leukocyte antigen-DR (HLA-DR) on monocytes provide greater insights into immune function. Low levels of monocyte HLA-DR (mHLA-DR) indicate immune suppression and have been linked to poor outcomes in patients experiencing septic shock. This finding suggests a potential role for immunostimulatory therapies, such as interferon-gamma [1,62,63]. Recent advancements in sepsis research have resulted in the identification of distinct phenotypes based on clinical and biomarker profiles. These sepsis phenotypes facilitate the stratification of patients into subgroups with varying responses to therapy. The Sepsis-3 definition underscores the importance of biomarkers in enhancing sepsis phenotyping beyond traditional classifications [15]. Biomarkers like lactate, PCT, and mHLA-DR are useful for distinguishing different sepsis phenotypes and may assist in developing precision medicine approaches for treating sepsis. Additionally, researchers are investigating transcriptomic and proteomic analyses to refine these phenotypic classifications even further [64]. To further illustrate the clinical applicability of biomarkers, Table 3 summarizes their potential utility across different stages of sepsis care, from prehospital assessment to post-acute follow-up. This stage-based approach emphasizes that no single biomarker is universally applicable; instead, their value depends on the timing and the clinical setting. Among shock-related biomarkers, lactate remains the most validated and guideline-endorsed tool, providing prognostic information and serving as a resuscitation target. However, lactate is not sepsis-specific and may be elevated in other conditions such as seizures, hepatic dysfunction, or β-adrenergic stimulation, which can confound

interpretation. Renin has recently gained attention as a marker of vasodilatory shock severity, with some studies suggesting prognostic superiority over lactate in predicting outcomes [42,65,66]. Nevertheless, renin does not reflect infection per se, requires specialized assays that are not routinely available, and its thresholds for clinical decision-making are still undefined [44,48,49,67]. The use of biomarker-based phenotyping of septic shock is attractive for stratifying patients and tailoring therapies, but current classifications remain exploratory and have not yet translated into routine practice. Overall, while lactate continues to serve as a pragmatic bedside marker, emerging candidates such as renin should be considered hypothesis-generating tools rather than established clinical standards.

Table 3. Biomarker utility across different clinical stages of sepsis.

Clinical Stage	Main Biomarkers	Clinical Purpose	Strengths	Limitations
Prehospital/Emergency	CRP, PCT, Lactate, PSP	Early identification of infection; risk stratification; triage	Rapid tests available; PSP shows early rise	Limited specificity; resource-dependent
Hospital Admission (ED/Ward)	PCT, CRP, Lactate, IL-6	Confirm infection vs. non-infectious inflammation; assess severity	Widely available; lactate recommended in guidelines	IL-6 non-specific; CRP slow kinetics
ICU/Septic Shock	Lactate, PCT, Renin, mHLA-DR, NGAL	Assess shock severity; prognosis; immune status; AKI prediction	Lactate clearance validated; renin and mHLA-DR promising; NGAL sensitive for AKI	Renin/NGAL not infection-specific; mHLA-DR requires flow cytometry
Post-acute/Recovery	mHLA-DR, CRP, PCT (trend)	Monitor immune recovery; risk of secondary infections; guide antibiotic discontinuation	Trends useful; immune monitoring possible	Limited validation; thresholds uncertain

3.2. Biomarker for Earlier Discontinuation of Antibiotics

Biomarkers play a crucial role in distinguishing between infections and non-infections, aiding in the decision to discontinue antibiotics early. This dual function is especially important in emergency and critical care settings, where it can be difficult to determine the underlying causes of systemic inflammation. By accurately identifying the presence of an infection, biomarkers contribute to effective antimicrobial stewardship efforts, as demonstrated by studies Wirz et al. and Arulkumaran et al. [68,69]. Christ-Crain et al. in the 2004 demonstrated that procalcitonin (PCT) levels could safely support the decision not to initiate antibiotics in patients presenting to the emergency department with suspected lower respiratory tract infections, thereby reducing unnecessary antibiotic use without compromising patient outcomes [5]. This "rule-out" capacity of PCT has since been replicated in multiple clinical settings, reinforcing its role in helping clinicians identify patients in whom a bacterial infection is unlikely [70,71].

Beyond the initial decision to start antibiotics, biomarkers are increasingly used to support early antibiotic discontinuation once infection is controlled. Several large-scale trials and meta-analyses, including those by de Jong et al. and Schuetz et al., have shown that PCT-guided strategies can significantly shorten antibiotic duration in sepsis without increasing mortality, and in some cases, even improving clinical outcomes [72–74]. These protocols rely on serial measurements, with predefined thresholds indicating when infection resolution is likely. However, PCT is not without limitations—its kinetics may be influenced by renal dysfunction, surgery, or trauma, and access to the assay remains inconsistent across centers [75].

Other biomarkers, such as C-reactive protein (CRP) and interleukin-6 (IL-6), have been investigated in this context. While CRP is widely available and inexpensive, its longer

half-life and slower kinetics make it less suitable for real-time decision-making. Although IL-6 is a rapid-response cytokine, it lacks specificity and is now considered suboptimal as a standalone marker for assessing infection resolution [76]. Biomarkers like presepsin and soluble TREM-1 have shown promise in early studies, but require further validation before being integrated into routine clinical algorithms [77,78]. Of note, the combination of biomarkers such as PCT with CRP may improve diagnostic accuracy and facilitate safer, earlier discontinuation of antibiotics [79].

These findings underscore the role of infection biomarkers not only in identifying patients who may not require antibiotics, but also in supporting the timely discontinuation of therapy once clinical improvement is evident. Strong evidence supports PCT-based algorithms, yet their implementation remains uneven due to cost, availability, and variability in clinician confidence. Future research should aim to standardize biomarker use across care settings, refine interpretation thresholds, and evaluate multimarker approaches that may enhance diagnostic and prognostic accuracy [80–82].

3.3. Biomarkers for Post-Acute Sepsis and Recurrence Prevention

During the post-acute phase of sepsis, the biomarkers are the same; often, their clinical trends have more significance than a single value itself. The most widely studied biomarker in post-acute sepsis is procalcitonin: the serial PCT measurements can help differentiate between true sepsis recurrence and other non-infectious inflammation causes, such as post-septic immune reconstitution syndrome. However, PCT should always be interpreted alongside clinical findings, as it may remain elevated due to non-infectious inflammatory conditions [83]. The Cochrane group emphasized that additional studies are needed to verify this result as also confirmed by Schuetz et al. and Papp et al. [84,85]. The CRP is commonly used in clinical practice, but its levels during sepsis resolution have been linked to poor outcomes, prolonged immune activation, and an increased risk of secondary infections. Although CRP is a non-specific marker of inflammation, it remains valuable in tracking post-sepsis recovery and identifying patients at risk for recurrence [86]. The combination of these biomarkers with the most recent molecules described likely offers higher accuracy in cases of antibiotic discontinuation or sepsis recurrence [87]. Although there are challenges, biomarkers remain clinically significant, particularly in low-income countries where modern technologies may be lacking. Sepsis is a major cause of mortality in vulnerable populations, including neonates, children, and the elderly. In these countries, preventing the recurrence of sepsis could greatly reduce mortality rates and positively impact the healthcare system. Therefore, it is crucial to monitor and manage sepsis recurrence effectively [88,89].

4. Conclusions

This narrative review highlights that no single biomarker is universally applicable in sepsis. Classical markers such as procalcitonin, C-reactive protein, and lactate remain widely used but have significant limitations in specificity and kinetics. Emerging biomarkers including pancreatic stone protein, NGAL, and mHLA-DR provide valuable insights into infection dynamics, renal injury, and immune suppression, yet their integration into routine care is hampered by heterogeneous evidence, lack of standardized thresholds, and technical constraints. Renin is promising for shock severity but is not infection-specific. Overall, biomarkers should be considered complementary tools that enrich but do not replace clinical judgment, microbiology, and organ function assessment.

4.1. Limitations

This work is a narrative review; therefore, it does not follow the structured methodology of a systematic review and may be subject to selection and reporting bias. Only English-language studies were considered, and heterogeneity in study design, assay methods, and patient populations prevents quantitative synthesis. Furthermore, thresholds for clinical application are inconsistent, limiting generalizability.

4.2. Future Directions

Future research should prioritize the validation and standardization of novel biomarkers such as PSP and mHLA-DR across multicenter cohorts, the development of biomarker panels rather than single-marker strategies, and the integration of point-of-care platforms across emergency, ICU, and post-acute pathways. Trials should also address the cost-effectiveness, implementation outcomes, and clinical impact of biomarker-guided algorithms, with a focus on antimicrobial stewardship, immune monitoring, and shock phenotyping [82,90].

Author Contributions: Conceptualization, G.M. and F.A.; methodology, F.G. and E.B.; software, M.V.; validation, E.B. and A.B.; formal analysis F.G.; investigation, G.M.; resources, data curation F.A.; writing—original draft preparation G.M. and F.A.; writing—review and editing, F.G.; visualization; supervision, E.B.; project administration, A.B. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Not applicable.

Acknowledgments: The authors acknowledge the use the help of AI tools for editing and improving the English language in their work.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Giamarellos-Bourboulis, E.J.; Aschenbrenner, A.C.; Bauer, M.; Bock, C.; Calandra, T.; Gat-Viks, I.; Kyriazopoulou, E.; Lupse, M.; Monneret, G.; Pickkers, P.; et al. The pathophysiology of sepsis and precision-medicine-based immunotherapy. *Nat. Immunol.* **2024**, *25*, 19–28. [CrossRef]
2. Cajander, S.; Kox, M.; Scicluna, B.P.; Weigand, M.A.; Mora, R.A.; Flohé, S.B.; Martin-Loeches, I.; Lachmann, G.; Girardis, M.; Garcia-Salido, A.; et al. Profiling the dysregulated immune response in sepsis: Overcoming challenges to achieve the goal of precision medicine. *Lancet Respir. Med.* **2024**, *12*, 305–322. [CrossRef]
3. Rhodes, A.; Evans, L.E.; Alhazzani, W.; Levy, M.M.; Antonelli, M.; Ferrer, R.; Kumar, A.; Sevransky, J.E.; Sprung, C.L.; Nunnally, M.E.; et al. Surviving Sepsis Campaign: International Guidelines for Management of Sepsis and Septic Shock: 2016. *Intensive Care Med.* **2017**, *43*, 304–377. [CrossRef]
4. Singer, M.; Deutschman, C.S.; Seymour, C.W.; Shankar-Hari, M.; Annane, D.; Bauer, M.; Bellomo, R.; Bernard, G.R.; Chiche, J.D.; Coopersmith, C.M.; et al. The Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3). *JAMA* **2016**, *315*, 801–810. [CrossRef]
5. Christ-Crain, M.; Jaccard-Stolz, D.; Bingisser, R.; Gencay, M.M.; Huber, P.R.; Tamm, M.; Müller, B. Effect of procalcitonin-guided treatment on antibiotic use and outcome in lower respiratory tract infections: Cluster-randomised, single-blinded intervention trial. *Lancet* **2004**, *363*, 600–607. [CrossRef]
6. Schlapbach, L.J.; Watson, R.S.; Sorce, L.R.; Argent, A.C.; Menon, K.; Hall, M.W.; Akech, S.; Albers, D.J.; Alpern, E.R.; Balamuth, F.; et al. International Consensus Criteria for Pediatric Sepsis and Septic Shock. *JAMA* **2024**, *331*, 665–674. [CrossRef]
7. Schuetz, P.; Christ-Crain, M.; Wolbers, M.; Schild, U.; Thomann, R.; Falconnier, C.; Widmer, I.; Neidert, S.; Blum, C.A.; Schönenberger, R.; et al. Procalcitonin guided antibiotic therapy and hospitalization in patients with lower respiratory tract infections: A prospective, multicenter, randomized controlled trial. *BMC Health Serv. Res.* **2007**, *7*, 102. [CrossRef] [PubMed]

8. Slim, M.A.; van Mourik, N.; Bakkerus, L.; Fuller, K.; Acharya, L.; Giannidis, T.; Dionne, J.C.; Oczkowski, S.J.W.; Netea, M.G.; Pickkers, P.; et al. Towards personalized medicine: A scoping review of immunotherapy in sepsis. *Crit. Care* **2024**, *28*, 183. [CrossRef] [PubMed]
9. Niederman, M.S.; Baron, R.M.; Bouadma, L.; Calandra, T.; Daneman, N.; DeWaele, J.; Kollef, M.H.; Lipman, J.; Nair, G.B. Initial antimicrobial management of sepsis. *Crit. Care* **2021**, *25*, 307. [CrossRef]
10. Baethge, C.; Goldbeck-Wood, S.; Mertens, S. SANRA-a scale for the quality assessment of narrative review articles. *Res. Integr. Peer Rev.* **2019**, *4*, 5. [CrossRef] [PubMed]
11. Aronson, J.K.; Ferner, R.E. Biomarkers-A General Review. *Curr. Protoc. Pharmacol.* **2017**, *76*, 9–23. [CrossRef]
12. Sprung, C.L.; Reinhart, K. Definitions for Sepsis and Septic Shock. *JAMA* **2016**, *316*, 456–457. [CrossRef]
13. Chimenti, C.; Sears, G.; McIntyre, J. Sepsis in Home Health Care: Screening, Education, and Rapid Triage. *J. Nurs. Care Qual.* **2021**, *36*, 210–216. [CrossRef] [PubMed]
14. Seymour, C.W.; Kennedy, J.N.; Wang, S.; Chang, C.H.; Elliott, C.F.; Xu, Z.; Berry, S.; Clermont, G.; Cooper, G.; Gomez, H.; et al. Derivation, Validation, and Potential Treatment Implications of Novel Clinical Phenotypes for Sepsis. *JAMA* **2019**, *321*, 2003–2017. [CrossRef]
15. Evans, L.; Rhodes, A.; Alhazzani, W.; Antonelli, M.; Coopersmith, C.M.; French, C.; Machado, F.R.; McIntyre, L.; Ostermann, M.; Prescott, H.C.; et al. Surviving sepsis campaign: International guidelines for management of sepsis and septic shock 2021. *Intensive Care Med.* **2021**, *47*, 1181–1247. [CrossRef] [PubMed]
16. Menon, K.; Schlapbach, L.J.; Akech, S.; Argent, A.; Biban, P.; Carrol, E.D.; Chiotos, K.; Jobayer Chisti, M.; Evans, I.V.R.; Inwald, D.P.; et al. Criteria for Pediatric Sepsis-A Systematic Review and Meta-Analysis by the Pediatric Sepsis Definition Taskforce. *Crit. Care Med.* **2022**, *50*, 21–36. [CrossRef] [PubMed]
17. Munroe, E.S.; Hyzy, R.C.; Semler, M.W.; Shankar-Hari, M.; Young, P.J.; Zampieri, F.G.; Prescott, H.C. Evolving Management Practices for Early Sepsis-induced Hypoperfusion: A Narrative Review. *Am. J. Respir. Crit. Care Med.* **2023**, *207*, 1283–1299. [CrossRef]
18. Bolanaki, M.; Winning, J.; Slagman, A.; Lehmann, T.; Kiehntopf, M.; Stacke, A.; Neumann, C.; Reinhart, K.; Möckel, M.; Bauer, M. Biomarkers Improve Diagnostics of Sepsis in Adult Patients With Suspected Organ Dysfunction Based on the Quick Sepsis-Related Organ Failure Assessment (qSOFA) Score in the Emergency Department. *Crit. Care Med.* **2024**, *52*, 887–899. [CrossRef]
19. Kim, T.H.; Kang, J.; Jang, H.; Joo, H.; Lee, G.Y.; Kim, H.; Cho, U.; Bang, H.; Jang, J.; Han, S.; et al. Blood culture-free ultra-rapid antimicrobial susceptibility testing. *Nature* **2024**, *632*, 893–902. [CrossRef]
20. Liang, J.; Cai, Y.; Shao, Y. Comparison of presepsin and Mid-regional pro-adrenomedullin in the diagnosis of sepsis or septic shock: A systematic review and meta-analysis. *BMC Infect. Dis.* **2023**, *23*, 288. [CrossRef]
21. Parri, N.; Trippella, G.; Lisi, C.; De Martino, M.; Galli, L.; Chiappini, E. Accuracy of presepsin in neonatal sepsis: Systematic review and meta-analysis. *Expert. Rev. Anti-Infect. Ther.* **2019**, *17*, 223–232. [CrossRef] [PubMed]
22. Yoon, S.H.; Kim, E.H.; Kim, H.Y.; Ahn, J.G. Presepsin as a diagnostic marker of sepsis in children and adolescents: A systemic review and meta-analysis. *BMC Infect. Dis.* **2019**, *19*, 760. [CrossRef]
23. Kang, S.; Kishimoto, T. Interplay between interleukin-6 signaling and the vascular endothelium in cytokine storms. *Exp. Mol. Med.* **2021**, *53*, 1116–1123. [CrossRef]
24. Zhao, J.O.; Patel, B.K.; Krishack, P.; Stutz, M.R.; Pearson, S.D.; Lin, J.; Lecompte-Osorio, P.A.; Dugan, K.C.; Kim, S.; Gras, N.; et al. Identification of Clinically Significant Cytokine Signature Clusters in Patients With Septic Shock. *Crit. Care Med.* **2023**, *51*, e253–e263. [CrossRef]
25. Gharamti, A.; Samara, O.; Monzon, A.; Scherger, S.; DeSanto, K.; Sillau, S.; Franco-Paredes, C.; Henao-Martínez, A.; Shapiro, L. Association between cytokine levels, sepsis severity and clinical outcomes in sepsis: A quantitative systematic review protocol. *BMJ Open* **2021**, *11*, e048476. [CrossRef]
26. Procházka, V.; Lacina, L.; Smetana, K., Jr.; Svoboda, M.; Skřivanová, K.; Beňovská, M.; Jarkovský, J.; Křen, L.; Kala, Z. Serum concentrations of proinflammatory biomarker interleukin-6 (IL-6) as a predictor of postoperative complications after elective colorectal surgery. *World J. Surg. Oncol.* **2023**, *21*, 384. [CrossRef]
27. Ooi, S.Z.Y.; Spencer, R.J.; Hodgson, M.; Mehta, S.; Phillips, N.L.; Preest, G.; Manivannan, S.; Wise, M.P.; Galea, J.; Zaben, M. Interleukin-6 as a prognostic biomarker of clinical outcomes after traumatic brain injury: A systematic review. *Neurosurg. Rev.* **2022**, *45*, 3035–3054. [CrossRef]
28. Pierrakos, C.; Vincent, J.L. Sepsis biomarkers: A review. *Crit. Care* **2010**, *14*, R15. [CrossRef]
29. Galvan, D.D.; Yu, Q. Surface-Enhanced Raman Scattering for Rapid Detection and Characterization of Antibiotic-Resistant Bacteria. *Adv. Healthc. Mater.* **2018**, *7*, e1701335. [CrossRef] [PubMed]
30. Klein, H.J.; Csordas, A.; Falk, V.; Slankamenac, K.; Rudiger, A.; Schönrath, F.; Rodriguez Cetina Biefer, H.; Starck, C.T.; Graf, R. Pancreatic stone protein predicts postoperative infection in cardiac surgery patients irrespective of cardiopulmonary bypass or surgical technique. *PLoS ONE* **2015**, *10*, e0120276. [CrossRef] [PubMed]

31. Klein, H.J.; Niggemann, P.; Buehler, P.K.; Lehner, F.; Schweizer, R.; Rittirsch, D.; Fuchs, N.; Waldner, M.; Steiger, P.; Giovanoli, P.; et al. Pancreatic Stone Protein Predicts Sepsis in Severely Burned Patients Irrespective of Trauma Severity: A Monocentric Observational Study. *Ann. Surg.* **2021**, *274*, e1179–e1186. [CrossRef] [PubMed]
32. Fidalgo, P.; Nora, D.; Coelho, L.; Povoa, P. Pancreatic Stone Protein: Review of a New Biomarker in Sepsis. *J. Clin. Med.* **2022**, *11*, 1085. [CrossRef] [PubMed]
33. Bradley, Z.; Bhalla, N. Point-of-care diagnostics for sepsis using clinical biomarkers and microfluidic technology. *Biosens. Bioelectron.* **2023**, *227*, 115181. [CrossRef]
34. Melegari, G.; Giuliani, E.; Di Pietro, G.; Alberti, F.; Campitiello, M.; Bertellini, E.; Barbieri, A. Point-of-care pancreatic stone protein measurement in critically ill COVID-19 patients. *BMC Anesthesiol.* **2023**, *23*, 226. [CrossRef]
35. Ashley, B.K.; Hassan, U. Point-of-critical-care diagnostics for sepsis enabled by multiplexed micro and nanosensing technologies. *Wiley Interdiscip. Rev. Nanomed. Nanobiotechnol.* **2021**, *13*, e1701. [CrossRef]
36. Kuil, S.D.; Hidad, S.; Fischer, J.C.; Harting, J.; Hertogh, C.M.; Prins, J.M.; van Leth, F.; de Jong, M.D.; Schneeberger, C. Sensitivity of point-of-care testing C reactive protein and procalcitonin to diagnose urinary tract infections in Dutch nursing homes: PROGRESS study protocol. *BMJ Open* **2019**, *9*, e031269. [CrossRef]
37. Bottari, G.; Paionni, E.; Fegatelli, D.A.; Murciano, M.; Rosati, F.; Ferrigno, F.; Pisani, M.; Cristaldi, S.; Musolino, A.; Borrelli, G.; et al. Pancreatic Stone Protein in the Diagnosis of Sepsis in Children Admitted to High-Dependency Care: A Single-Center Prospective Cohort Study. *Pediatr. Crit. Care Med.* **2024**, *25*, 937–946. [CrossRef]
38. Yang, H.; Du, L.; Zhang, Z. Potential biomarkers in septic shock besides lactate. *Exp. Biol. Med.* **2020**, *245*, 1066–1072. [CrossRef] [PubMed]
39. Bakker, J.; Gris, P.; Coffernils, M.; Kahn, R.J.; Vincent, J.L. Serial blood lactate levels can predict the development of multiple organ failure following septic shock. *Am. J. Surg.* **1996**, *171*, 221–226. [CrossRef]
40. Vincent, J.L.; Quintairos, E.S.A.; Couto, L., Jr.; Taccone, F.S. The value of blood lactate kinetics in critically ill patients: A systematic review. *Crit. Care* **2016**, *20*, 257. [CrossRef]
41. Alshiakh, S.M. Role of serum lactate as prognostic marker of mortality among emergency department patients with multiple conditions: A systematic review. *SAGE Open Med.* **2023**, *11*, 20503121221136401. [CrossRef] [PubMed]
42. Khodashahi, R.; Sarjamee, S. Early lactate area scores and serial blood lactate levels as prognostic markers for patients with septic shock: A systematic review. *Infect. Dis.* **2020**, *52*, 451–463. [CrossRef] [PubMed]
43. Bellomo, R.; Forni, L.G.; Busse, L.W.; McCurdy, M.T.; Ham, K.R.; Boldt, D.W.; Hästbacka, J.; Khanna, A.K.; Albertson, T.E.; Tumlin, J.; et al. Renin and Survival in Patients Given Angiotensin II for Catecholamine-Resistant Vasodilatory Shock. A Clinical Trial. *Am. J. Respir. Crit. Care Med.* **2020**, *202*, 1253–1261. [CrossRef]
44. Legrand, M.; Khanna, A.K.; Ostermann, M.; Kotani, Y.; Ferrer, R.; Girardis, M.; Leone, M.; DePascale, G.; Pickkers, P.; Tissieres, P.; et al. The renin-angiotensin-aldosterone-system in sepsis and its clinical modulation with exogenous angiotensin II. *Crit. Care* **2024**, *28*, 389. [CrossRef]
45. Hilgenfeldt, U.; Kienapfel, G.; Kellermann, W.; Schott, R.; Schmidt, M. Renin-angiotensin system in sepsis. *Clin. Exp. Hypertens. A* **1987**, *9*, 1493–1504. [CrossRef]
46. Barbieri, A.; Giuliani, E.; Marchetti, G.; Ugoletti, E.; Della Volpe, S.; Albertini, G. Plasma renin concentration as a predictor of outcome in a medical intensive care setting: A retrospective pilot study. *Minerva Anestesiol.* **2012**, *78*, 1248–1253.
47. Gleeson, P.J.; Crippa, I.A.; Mongkolpun, W.; Cavicchi, F.Z.; Van Meerhaeghe, T.; Brimioulle, S.; Taccone, F.S.; Vincent, J.L.; Creteur, J. Renin as a Marker of Tissue-Perfusion and Prognosis in Critically Ill Patients. *Crit. Care Med.* **2019**, *47*, 152–158. [CrossRef]
48. Khanna, A.K. Tissue Perfusion and Prognosis in the Critically Ill-Is Renin the New Lactate? *Crit. Care Med.* **2019**, *47*, 288–290. [CrossRef]
49. Jeyaraju, M.; McCurdy, M.T.; Levine, A.R.; Devarajan, P.; Mazzeffi, M.A.; Mullins, K.E.; Reif, M.; Yim, D.N.; Parrino, C.; Lankford, A.S.; et al. Renin Kinetics Are Superior to Lactate Kinetics for Predicting In-Hospital Mortality in Hypotensive Critically Ill Patients. *Crit. Care Med.* **2022**, *50*, 50–60. [CrossRef]
50. Khanna, A.; English, S.W.; Wang, X.S.; Ham, K.; Tumlin, J.; Szerlip, H.; Busse, L.W.; Altaweel, L.; Albertson, T.E.; Mackey, C.; et al. Angiotensin II for the Treatment of Vasodilatory Shock. *N. Engl. J. Med.* **2017**, *377*, 419–430. [CrossRef] [PubMed]
51. Leone, M.; Einav, S.; Antonucci, E.; Depret, F.; Lakbar, I.; Martin-Loeches, I.; Wieruszewski, P.M.; Myatra, S.N.; Khanna, A.K. Multimodal strategy to counteract vasodilation in septic shock. *Anaesth. Crit. Care Pain. Med.* **2023**, *42*, 101193. [CrossRef]
52. Ammar, M.A.; Ammar, A.A.; Wieruszewski, P.M.; Bissell, B.D.; Long, M.T.; Albert, L.; Khanna, A.K.; Sacha, G.L. Timing of vasoactive agents and corticosteroid initiation in septic shock. *Ann. Intensive Care* **2022**, *12*, 47. [CrossRef]
53. Teixeira, J.P.; Perez Ingles, D.; Barton, J.B.; Dean, J.T.; Garcia, P.; Kunkel, S.J.; Sarangarm, P.; Weiss, N.K.; Schaich, C.L.; Busse, L.W.; et al. The scientific rationale and study protocol for the DPP3, Angiotensin II, and Renin Kinetics in Sepsis (DARK-Sepsis) randomized controlled trial: Serum biomarkers to predict response to angiotensin II versus standard-of-care vasopressor therapy in the treatment of septic shock. *Trials* **2024**, *25*, 182. [CrossRef] [PubMed]

54. Flannery, A.H.; Kiser, A.S.; Behal, M.L.; Li, X.; Neyra, J.A. RAS inhibition and sepsis-associated acute kidney injury. *J. Crit. Care* **2022**, *69*, 153986. [CrossRef] [PubMed]
55. Tibi, S.; Zeynalvand, G.; Mohsin, H. Role of the Renin Angiotensin Aldosterone System in the Pathogenesis of Sepsis-Induced Acute Kidney Injury: A Systematic Review. *J. Clin. Med.* **2023**, *12*, 4566. [CrossRef]
56. Salgado, D.R.; Rocco, J.R.; Silva, E.; Vincent, J.L. Modulation of the renin-angiotensin-aldosterone system in sepsis: A new therapeutic approach? *Expert. Opin. Ther. Targets* **2010**, *14*, 11–20. [CrossRef] [PubMed]
57. Ferreira, G.S.; Frota, M.L.; Gonzaga, M.J.D.; Vattimo, M.F.F.; Lima, C. The Role of Biomarkers in Diagnosis of Sepsis and Acute Kidney Injury. *Biomedicines* **2024**, *12*, 931. [CrossRef]
58. Hoste, E.A.; Bagshaw, S.M.; Bellomo, R.; Cely, C.M.; Colman, R.; Cruz, D.N.; Edipidis, K.; Forni, L.G.; Gomersall, C.D.; Govil, D.; et al. Epidemiology of acute kidney injury in critically ill patients: The multinational AKI-EPI study. *Intensive Care Med.* **2015**, *41*, 1411–1423. [CrossRef]
59. Singh, R.; Watchorn, J.C.; Zarbock, A.; Forni, L.G. Prognostic Biomarkers and AKI: Potential to Enhance the Identification of Post-Operative Patients at Risk of Loss of Renal Function. *Res. Rep. Urol.* **2024**, *16*, 65–78. [CrossRef]
60. Vijayan, A.; Faubel, S.; Askenazi, D.J.; Cerda, J.; Fissell, W.H.; Heung, M.; Humphreys, B.D.; Koyner, J.L.; Liu, K.D.; Mour, G.; et al. Clinical Use of the Urine Biomarker [TIMP-2] × [IGFBP7] for Acute Kidney Injury Risk Assessment. *Am. J. Kidney Dis.* **2016**, *68*, 19–28. [CrossRef]
61. Monneret, G.; Venet, F. Monocyte HLA-DR in sepsis: Shall we stop following the flow? *Crit. Care* **2014**, *18*, 102. [CrossRef]
62. van der Poll, T.; Shankar-Hari, M.; Wiersinga, W.J. The immunology of sepsis. *Immunity* **2021**, *54*, 2450–2464. [CrossRef]
63. Shankar-Hari, M.; Calandra, T.; Soares, M.P.; Bauer, M.; Wiersinga, W.J.; Prescott, H.C.; Knight, J.C.; Baillie, K.J.; Bos, L.D.J.; Derde, L.P.G.; et al. Reframing sepsis immunobiology for translation: Towards informative subtyping and targeted immunomodulatory therapies. *Lancet Respir. Med.* **2024**, *12*, 323–336. [CrossRef]
64. Chen, H.; Luo, H.; Tian, T.; Li, S.; Jiang, Y. Integrated Analyses of Single-Cell Transcriptome and Mendelian Randomization Reveal the Protective Role of Resistin in Sepsis Survival in Intensive Care Unit. *Int. J. Mol. Sci.* **2023**, *24*, 14982. [CrossRef] [PubMed]
65. Houwink, A.P.; Rijkenberg, S.; Bosman, R.J.; van der Voort, P.H. The association between lactate, mean arterial pressure, central venous oxygen saturation and peripheral temperature and mortality in severe sepsis: A retrospective cohort analysis. *Crit. Care* **2016**, *20*, 56. [CrossRef] [PubMed]
66. Brooks, G.A. The Science and Translation of Lactate Shuttle Theory. *Cell Metab.* **2018**, *27*, 757–785. [CrossRef] [PubMed]
67. Kotani, Y.; Belletti, A.; Maiucci, G.; Lodovici, M.; Fresilli, S.; Landoni, G.; Bellomo, R.; Zarbock, A. Renin as a Prognostic Marker in Intensive Care and Perioperative Settings: A Scoping Review. *Anesth. Analg.* **2024**, *138*, 929–936. [CrossRef]
68. Wirz, Y.; Meier, M.A.; Bouadma, L.; Luyt, C.E.; Wolff, M.; Chastre, J.; Tubach, F.; Schroeder, S.; Nobre, V.; Annane, D.; et al. Effect of procalcitonin-guided antibiotic treatment on clinical outcomes in intensive care unit patients with infection and sepsis patients: A patient-level meta-analysis of randomized trials. *Crit. Care* **2018**, *22*, 191. [CrossRef]
69. Arulkumaran, N.; Khpal, M.; Tam, K.; Baheerathan, A.; Corredor, C.; Singer, M. Effect of Antibiotic Discontinuation Strategies on Mortality and Infectious Complications in Critically Ill Septic Patients: A Meta-Analysis and Trial Sequential Analysis. *Crit. Care Med.* **2020**, *48*, 757–764. [CrossRef]
70. Prkno, A.; Wacker, C.; Brunkhorst, F.M.; Schlattmann, P. Procalcitonin-guided therapy in intensive care unit patients with severe sepsis and septic shock--a systematic review and meta-analysis. *Crit. Care* **2013**, *17*, R291. [CrossRef]
71. Gu, W.J.; Liu, J.C. Procalcitonin-guided therapy in severe sepsis and septic shock. *Crit. Care* **2014**, *18*, 427. [CrossRef]
72. de Jong, E.; van Oers, J.A.; Beishuizen, A.; Vos, P.; Vermeijden, W.J.; Haas, L.E.; Loef, B.G.; Dormans, T.; van Melsen, G.C.; Kluiters, Y.C.; et al. Efficacy and safety of procalcitonin guidance in reducing the duration of antibiotic treatment in critically ill patients: A randomised, controlled, open-label trial. *Lancet Infect. Dis.* **2016**, *16*, 819–827. [CrossRef]
73. Schuetz, P.; Beishuizen, A.; Broyles, M.; Ferrer, R.; Gavazzi, G.; Gluck, E.H.; González Del Castillo, J.; Jensen, J.U.; Kanizsai, P.L.; Kwa, A.L.H.; et al. Procalcitonin (PCT)-guided antibiotic stewardship: An international experts consensus on optimized clinical use. *Clin. Chem. Lab. Med.* **2019**, *57*, 1308–1318. [CrossRef]
74. Schuetz, P.; Wirz, Y.; Sager, R.; Christ-Crain, M.; Stolz, D.; Tamm, M.; Bouadma, L.; Luyt, C.E.; Wolff, M.; Chastre, J.; et al. Effect of procalcitonin-guided antibiotic treatment on mortality in acute respiratory infections: A patient level meta-analysis. *Lancet Infect. Dis.* **2018**, *18*, 95–107. [CrossRef]
75. Gregoriano, C.; Heilmann, E.; Molitor, A.; Schuetz, P. Role of procalcitonin use in the management of sepsis. *J. Thorac. Dis.* **2020**, *12*, S5–S15. [CrossRef]
76. Hung, S.K.; Lan, H.M.; Han, S.T.; Wu, C.C.; Chen, K.F. Current Evidence and Limitation of Biomarkers for Detecting Sepsis and Systemic Infection. *Biomedicines* **2020**, *8*, 494. [CrossRef]
77. Masson, S.; Caironi, P.; Spanuth, E.; Thomae, R.; Panigada, M.; Sangiorgi, G.; Fumagalli, R.; Mauri, T.; Isgrò, S.; Fanizza, C.; et al. Presepsin (soluble CD14 subtype) and procalcitonin levels for mortality prediction in sepsis: Data from the Albumin Italian Outcome Sepsis trial. *Crit. Care* **2014**, *18*, R6. [CrossRef] [PubMed]

78. Kung, C.T.; Su, C.M.; Hsiao, S.Y.; Chen, F.C.; Lai, Y.R.; Huang, C.C.; Lu, C.H. The Prognostic Value of Serum Soluble TREM-1 on Outcome in Adult Patients with Sepsis. *Diagnostics* **2021**, *11*, 1979. [CrossRef]
79. Kyriazopoulou, E.; Giamarellos-Bourboulis, E.J. Antimicrobial Stewardship Using Biomarkers: Accumulating Evidence for the Critically Ill. *Antibiotics* **2022**, *11*, 367. [CrossRef] [PubMed]
80. Pierrakos, C.; Velissaris, D.; Bisdorff, M.; Marshall, J.C.; Vincent, J.L. Biomarkers of sepsis: Time for a reappraisal. *Crit. Care* **2020**, *24*, 287. [CrossRef] [PubMed]
81. Lee, E.H.; Lee, K.H.; Song, Y.G.; Han, S.H. Discrepancy of C-Reactive Protein, Procalcitonin and Interleukin-6 at Hospitalization: Infection in Patients with Normal C-Reactive Protein, Procalcitonin and High Interleukin-6 Values. *J. Clin. Med.* **2022**, *11*, 7324. [CrossRef]
82. Marshall, J.C.; Reinhart, K. Biomarkers of sepsis. *Crit. Care Med.* **2009**, *37*, 2290–2298. [CrossRef] [PubMed]
83. Kyriazopoulou, E.; Liaskou-Antoniou, L.; Adamis, G.; Panagaki, A.; Melachroinopoulos, N.; Drakou, E.; Marousis, K.; Chrysos, G.; Spyrou, A.; Alexiou, N.; et al. Procalcitonin to Reduce Long-Term Infection-associated Adverse Events in Sepsis. A Randomized Trial. *Am. J. Respir. Crit. Care Med.* **2021**, *203*, 202–210. [CrossRef] [PubMed]
84. Schuetz, P.; Wirz, Y.; Sager, R.; Christ-Crain, M.; Stolz, D.; Tamm, M.; Bouadma, L.; Luyt, C.E.; Wolff, M.; Chastre, J.; et al. Procalcitonin to initiate or discontinue antibiotics in acute respiratory tract infections. *Cochrane Database Syst. Rev.* **2017**, *10*, Cd007498. [CrossRef] [PubMed]
85. Papp, M.; Kiss, N.; Baka, M.; Trásy, D.; Zubek, L.; Fehérvári, P.; Harnos, A.; Turan, C.; Hegyi, P.; Molnár, Z. Procalcitonin-guided antibiotic therapy may shorten length of treatment and may improve survival-a systematic review and meta-analysis. *Crit. Care* **2023**, *27*, 394. [CrossRef]
86. Kubo, K.; Sakuraya, M.; Sugimoto, H.; Takahashi, N.; Kano, K.I.; Yoshimura, J.; Egi, M.; Kondo, Y. Benefits and Harms of Procalcitonin- or C-Reactive Protein-Guided Antimicrobial Discontinuation in Critically Ill Adults With Sepsis: A Systematic Review and Network Meta-Analysis. *Crit. Care Med.* **2024**, *52*, e522–e534. [CrossRef]
87. Njunge, J.M.; Gwela, A.; Kibinge, N.K.; Ngari, M.; Nyamako, L.; Nyatichi, E.; Thitiri, J.; Gonzales, G.B.; Bandsma, R.H.J.; Walson, J.L.; et al. Biomarkers of post-discharge mortality among children with complicated severe acute malnutrition. *Sci. Rep.* **2019**, *9*, 5981. [CrossRef]
88. Ishaque, S.; Famularo, S.T., 3rd; Saleem, A.F.; Siddiqui, N.U.R.; Kazi, Z.; Parkar, S.; Hotwani, A.; Thomas, N.J.; Thompson, J.M.; Lahni, P.; et al. Biomarker-Based Risk Stratification in Pediatric Sepsis From a Low-Middle Income Country. *Pediatr. Crit. Care Med.* **2023**, *24*, 563–573. [CrossRef]
89. Mount, M.C.; Remy, K.E. Help Wanted for Sepsis: Biomarkers in Low- and Middle-Income Countries Please Apply. *Pediatr. Crit. Care Med.* **2023**, *24*, 619–621. [CrossRef]
90. Levy, M.M. The electrocardiogram for sepsis: How close are we? *Crit. Care* **2007**, *11*, 144. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.